TENNIS BUILD SPECIFICATION

Separate repository. Separate database. No shared runtime with the football system.

Read `00-developer-brief.md` first. The non-negotiable rules (R1–R10), the arithmetic library, the three measurement layers and the staking rule all apply here and are not repeated.

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1. Scope

Men only. No WTA.

Level	In scope	Notes
Grand Slam (men's singles)	Yes	Best-of-five; format handled explicitly
ATP Masters 1000	Yes	
ATP 500 / 250	Yes	
ATP Challenger Tour	Yes, with a wider threshold	Data is thinner; limits are tiny; prices move fast
ATP qualifying	Only where reliable data exists	
Futures / ITF	**No**	Data quality insufficient
WTA / women's	**No**	

Markets:

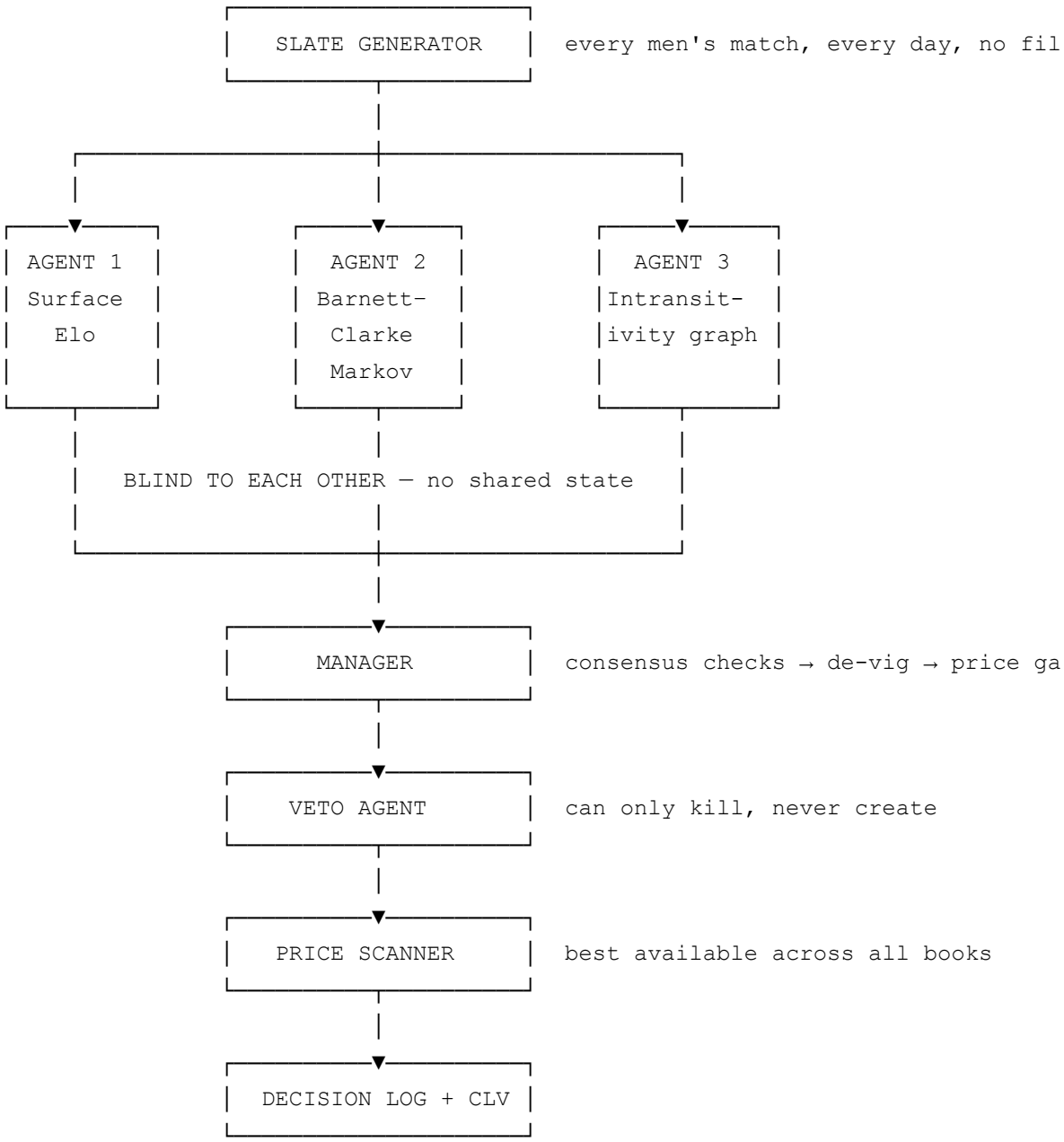
- Match winner (primary)
- Set betting and total games (secondary — see §6 on derivatives)
- Handicaps (games)

Not in scope: in-play, retirement settlement logic. We publish selections; we do not stake, so the differing retirement rules between bookmakers are the reader's problem to check,

not the system's to model. Pre-match injury and fatigue signals remain fully in scope as a veto.

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2. Architecture overview



"Blind" is a hard requirement. The three agents run in separate processes, receive only the fixture identifier and the permitted feature set, and must not be able to read each other's outputs or each other's intermediate state. If they can see one another, their errors correlate and the agreement rule stops carrying any information — you get three copies of one opinion wearing different hats.

3. Agent 1 — Surface-adjusted Elo

3.1 What it is

Elo is a rating that moves after every match: the winner takes points from the loser, and the amount transferred depends on how surprising the result was. Beating someone far below you moves your rating very little; beating someone far above you moves it a lot.

Maintain **four ratings per player**: overall, hard, clay, grass.

3.2 Mechanics

Standard Elo update:

```
E_A     = 1 / (1 + 10^((R_B − R_A)/400))        # expected score for A
R_A'    = R_A + K · (S_A − E_A)                 # S_A = 1 if A won, 0 if lost
```

Starting ratings: around 1500 at tour level; lower tiers start around 1200. Roughly 2000 separates the genuine elite — typically only a handful of men hold a rating that high at any one time.

K-factor. Use a decaying K so that a player's rating stabilises as their match count grows, rather than a fixed K. A common form is `K = a/(m + b)^c` where `m` is the player's career match count. Fit `a`, `b`, `c` on historical data by minimising Brier score — **and record that you did so as a fitted-variant count** (see R4 and §9 of the brief).

3.3 The surface blend — this specific number matters

```
R_blended = 0.5 · R_surface + 0.5 · R_overall
```

Fifty-fifty. Sackmann tested other weights and found 50/50 worked for every surface, which is a notably blunt result — no surface warrants more or less surface-specificity than any other. Do not "improve" this without a walk-forward test, and if you test alternatives, count them.

3.4 Output

Convert the blended rating difference to a match probability with the standard Elo expectation. For best-of-five, apply the format adjustment in §4.6 rather than using a separate rating.

3.5 What you must expect from this agent

Sackmann's own published comparison:

Model	Accuracy	Brier
Pinnacle (market)	76.92%	.172
FiveThirtyEight	75.21%	**.171**
Tennis Abstract (Elo-based)	74.36%	.182
ATP ranking	72.65%	.185

A well-built surface Elo will be worse-calibrated than the closing price. That is the expected result, not a bug in your implementation. Its job here is to be one of three independent probability estimates and to act as a sanity check on the other two — not to be a source of edge on its own.

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4. Agent 2 — Barnett–Clarke hierarchical Markov model

4.1 The idea, for someone who hasn't done this before

A tennis match result is a lossy compression of the point results. Two players who each win 55% of their matches can be completely different if one wins comfortably and the other scrapes tiebreaks. So we model at the **point** level and then propagate deterministically upward:

```
point → game → tiebreak → set → match
```

The whole model rests on **two numbers**: the probability each player wins a point on his own serve, against this specific opponent, on this surface. Everything else is arithmetic.

Why this is worth the effort over a single win-probability number:

1. **Scoring structure is violently non-linear.** A two-percentage-point serve edge compounds through the game/set/match hierarchy into a much larger match edge. A single probability cannot express that.
2. **It prices every derivative from one object.** Set betting, correct score, total games, handicaps and tiebreak lines all fall out of the same simulation.
3. **Format sensitivity is free.** Best-of-five favours the better player, because a larger

sample gives the stronger player more chance to realise his edge.

4. **It separates *who wins* from *how*.** Two matchups with identical 65% favourites imply completely different game totals depending on whether the 65% comes from serve dominance (short match, tiebreaks) or return dominance (long match, many breaks).

Known violation, stated honestly: points are not truly independent and identically distributed. Klaassen & Magnus is the canonical test. The deviations exist but are small, which is why the i.i.d. model survives as the working standard despite being known to be false. Do not build a correction for this; nobody has published a reliable one and you would be fitting noise.

4.2 Step one — raw serve statistics

From Sackmann's `tennis_atp` match files, per player per match:

- Service points won (`SPW`) = (first serves won + second serves won) / service points played
- Return points won (`RPW`) = 1 – opponent's SPW

Aggregate over a trailing window. Suggested: 12 months, minimum 20 matches, surface-weighted using the same 50/50 principle as Elo. Below the minimum, shrink toward the tour average for that surface — do not use a thin sample raw.

Benchmarks to sanity-check your numbers against: a top-100 player averages roughly 80% of service games won and 20% of return games won. A big-server archetype runs closer to 90% / 10%. ATP first-set tiebreak frequency is generally under 20%, but rises to around 43% in big-server-versus-big-server matchups.

4.3 Step two — the opponent adjustment

This is the single most important line of arithmetic in the tennis build:

```
p_A_serve = SPW_A − (RPW_B − RPW_tour_average)
```

In words: take player A's service-point win rate, then subtract the amount by which player B's return is better than the tour average. If B returns better than average, A's serve number comes down; if B returns worse than average, it goes up.

Without this, a player who has spent six months beating weak returners looks like a great server and a player grinding against elite returners looks mediocre. Raw SPW is contaminated by schedule.

The better parametric form, if you want to fit it properly:

```
P(server wins point) = (1 + 10^(-(θ + (S − R)/d)))^(-1)
```

where `S` is the server's rating, `R` the returner's rating, `d` a scaling parameter, and **`θ` is a surface-specific serve advantage**. The `θ` term is the important bit: without it, an average server against an average returner appears evenly matched, which is wrong on grass and wrong in the other direction on clay.

The interpretive trap you must not fall into: these are *points-won-on-serve* ratings, not serve-quality ratings. A famously huge server can rate below an all-court great on this measure, because the latter wins more service points through the rest of his game. Anyone reading "serve rating" as "quality of serve" will misprice every serve derivative.

4.4 Step three — the sum constraint

Force the two players' adjusted serve probabilities to sum to a fixed, surface-typical total:

```
p_A_serve + p_B_serve = t_surface
```

Rescale both to satisfy it. `t_surface` is calibrated from historical data per surface (it is higher on grass, lower on clay). This constraint stops the model producing nonsense like both men winning 90% of their service points, which the raw adjustment can easily generate when both players have favourable recent schedules.

4.5 Step four — propagate

Standard recursion. Let `p` be the server's point-win probability.

Game. Probability the server holds, from the score state `(a, b)`:

```
N(a, b) = p · N(a+1, b) + (1 − p) · N(a, b+1)
```

with terminal conditions at game won / lost, and deuce handled as `P(deuce) = p²/(p² + (1−p)²)`.

Tiebreak. Same recursion, alternating server per tiebreak serving order, to 7 points, win by two.

Set. Recursion over games with the two hold probabilities `p_A_hold` and `p_B_hold`, alternating serve, tiebreak at 6–6.

Match. Best-of-three: first to two sets. Best-of-five: first to three. Note the Grand Slam final-set rules differ by event (10-point tiebreak at 6–6 is now standard across all four, but verify per event per year in your historical data — this changed within the backtest window

and getting it wrong corrupts long-match pricing).

Implement the recursion with memoisation. It is exact — you do not need Monte Carlo for the match-winner market. Use simulation only for derivatives where the closed form is awkward.

4.6 Common-opponent routing — a component, not a strategy

When two players have few or no meetings and dissimilar schedules, route both through their shared opponents: compute each player's serve and return performance *against the specific set of opponents both have faced*, and compare on that common basis rather than against the global tour average.

Read this carefully before building it. The published common-opponent model reported +3.8% ROI over 2,173 ATP matches. A replication found that figure was driven by **a single outlier bet against erroneously long odds in the test data**. Cleaned up, it is not profitable in practical scenarios. The replicators concluded the residual return came from exploiting individual bookmakers' mispricing relative to the market — line shopping — not from the tennis model.

Therefore: common-opponent routing is **an input to `p_serve` estimation only**. It never generates a bet on its own, it is never presented as an edge, and no threshold is relaxed because of it.

Where it genuinely helps: Challenger level, where players share opponents constantly because they travel the same circuit, and where global tour averages are a poor reference. This is also the level where Sackmann's match-level data (serve points won, return points won per match) is available even though point-by-point charting is not — and match-level is sufficient for this model. You do not need point-by-point data to build it.

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5. Agent 3 — Intransitivity graph

5.1 What it detects

Tennis contains rock-paper-scissors relationships. A beats B, B beats C, and C beats A — because styles interact. A rating system that maps everyone onto a single line **cannot represent this by construction**, so it misprices exactly these matchups, and so, potentially, does the market.

5.2 Construction

Build a directed weighted graph:

- **Nodes** = players
- **Edges** = match results, directed loser → winner
- **Weights** = recency-decayed, optionally surface-partitioned

Detect cycles. For a given fixture, compute an **intransitivity score** measuring how much of the local graph structure around those two players cannot be explained by a linear ordering.

The published implementation uses a graph neural network over this structure to produce a match probability. arXiv 2510.20454 has the method.

5.3 The result, and why the filter is the whole point

Application	Return
Applied to all matches indiscriminately	**−5.49% ROI**
Applied only to high-intransitivity matchups	**+3.26% ROI**, Kelly staking, 1,903 bets

Reported accuracy on the selected subset was 65.7% with a Brier score of 0.215.

Build the filter first and treat the model as secondary. An intransitivity score that correctly identifies which matches are style-clash matches is the valuable object. Applied everywhere, this agent loses money by a wide margin.

5.4 Operational consequence

Agent 3 **abstains** on low-intransitivity fixtures rather than producing a low-confidence number. An abstention is a distinct output from a probability — see §6.2 for how the manager handles it.

Note also that this paper is recent (October 2025) and, as far as I can establish, has **not been independently replicated**. Given what happened to the common-opponent result, treat its +3.26% as a hypothesis your own walk-forward test must confirm, not as a number to rely on. Women's tennis shows around 11.5% more intransitivity than men's, which is out of our scope but worth knowing as context for why the men's effect is the smaller one.

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6. The manager

The manager is where the system stops predicting and starts pricing. **It votes on the price gap, not on who wins.** Three agents agreeing with the bookmaker is a fair-odds bet and a slow loss.

6.1 Inputs

For each fixture, three probabilities (or two plus an abstention), plus the full set of market prices from the price scanner.

6.2 Consensus rules

Applied in order. Any failure produces a logged no-bet with a reason code.

C1 — Directional agreement. All non-abstaining agents must favour the same player. Any disagreement on direction → **NO BET**.

C2 — Spread ceiling. The maximum minus the minimum of the non-abstaining probabilities must be **≤ 8 percentage points**. If the agents are 60%, 59% and 41%, that is a 19-point spread and the model is telling you it does not understand this match → **NO BET**.

C3 — Hard floor. No agent may have the selection below 50%. A single agent under 50% kills it regardless of the other two.

C4 — Abstention handling. If Agent 3 abstains, proceed with two agents but require the spread ceiling to be **≤ 6 points** rather than 8, since you have less to triangulate with. If two or more agents abstain or fail → **NO BET**.

Consensus probability = the mean of the non-abstaining agent probabilities. Do not weight them by historical performance in the first six months; you will not have enough data to weight on and you will simply be fitting noise. Revisit after 500 settled selections.

6.3 De-vig the reference price

1. Take the sharp reference price for both players (see §7.3 of the brief on obtaining it).
2. Compute booksum `S = 1/o_A + 1/o_B`.
3. De-vig using **power or Shin**. Match-winner tennis is a balanced two-way market so the methods will agree closely; use power as the default and record the method. On any market with a leg priced above 4.00 — set betting, correct score — power or Shin is mandatory rather than preferred.
4. Result: `p_market_fair` and `o_market_fair = 1/p_market_fair`.

6.4 The firing rule

```
edge = (p_consensus × best_available_price) − 1
```

Fire only if `edge ≥ threshold`.

Level	Threshold
ATP tour (250 / 500 / M1000 / Slam)	**5%**
ATP Challenger	**8%**
Any market with a leg priced above 4.00	**8%**

Why 5% and not 3%. A well-regarded public tennis tipster was measured beating Pinnacle's closing line by about 3% on average. Once Pinnacle's own margin is removed, that leaves roughly break-even. A 3% threshold therefore selects for bets that are, after costs, worth nothing. 5% is the minimum that clears the margin with something left over.

Why Challenger is higher. Thinner data, wider margins, faster-moving prices and tiny stake limits. Everything that makes an edge harder to realise argues for demanding a bigger one.

6.5 Minimum acceptable price

Every fired selection carries a `minimum_acceptable_price`, computed as the price at which the edge falls to the threshold:

```
min_price = (1 + threshold) / p_consensus
```

Publish it with the selection. Below that number the bet is dead, and anyone taking it after the price has moved is taking a different bet from the one the model made.

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7. The veto agent

The veto agent can only kill a selection. It can never create one, never raise a probability, and never lower a threshold.

The reasoning: general tennis knowledge — height, surface preference, nationality, serve quality — is already fully contained in the sharp price. Adding it as a fourth opinion would contribute noise, not information. What it *can* do is catch the specific, recent, discrete facts that a statistical model built on historical match data cannot see.

7.1 Checks

Check	Kills the bet if
Withdrawal	Player has withdrawn or retired from the event

Injury report	Credible injury or medical timeout report since the last completed match
Previous-match load	Long three-setter, or an unusually high total-games count, in the previous round with short turnaround
Turnaround time	Under a configured minimum hours between end of previous match and scheduled start
First match back	First event after a layoff of over a configured number of weeks
Retirement history	Elevated recent retirement rate flagged for manual review
Late schedule change	Court, time, or indoor/outdoor status changed after prediction
Data staleness	Any input feed outside its freshness window (see §5.3 of the brief)

7.2 What it must not do

- Must not consult the market price. It is not a second opinion on value.
- Must not un-veto based on the size of the edge. A big edge on an injured player is not a bet, it is a stale price.
- Must not produce a probability.

Every veto is logged with the trigger, so you can later measure whether the veto agent is adding value or just reducing volume.

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8. The price scanner

On the evidence, this is where the tennis money actually is. More books beats better models.

8.1 What it does

For each fixture on the slate, poll every available bookmaker for every in-scope market, on a schedule, storing every observation append-only in `odds_snapshots` with `observed_at`.

Then:

- Identify `best_available_price` per selection per market.

- Compare against `o_market_fair` from the sharp reference.
- Flag any book whose price exceeds the sharp fair price, regardless of whether the three agents fired.

8.2 The two independent signals

The system produces two kinds of flag, and they must be recorded distinctly:

Type A — model edge. The three agents agree, the consensus probability implies a fair price shorter than the best available price by at least the threshold. This is the "we think the market is wrong" bet.

Type B — price dispersion. A soft book's price exceeds the de-vigged sharp fair price by at least the threshold, with no reference to our own model at all. This is the "one bookmaker is out of step with the market" bet.

Type B is the better-evidenced of the two. Buchdahl measured **+3.40% yield over 22,281 bets** taking the best soft price where it beat Pinnacle's de-vigged fair odds, rising to **+13.22% over 1,927 bets** where the gap exceeded 5%. The control — deliberately betting below fair price — returned **–3.08% over 44,673 bets**.

Track them separately from day one. They will have different hit rates, different CLV profiles and different lifespans, and if they are mixed in one ledger you will never be able to tell which half is working.

8.3 Coverage is the priority

Twenty books beats five. Every additional bookmaker you can read multiplies the chance that one of them has lagged the market on a given fixture. This is data plumbing, not modelling, and it is the highest-return engineering work in the tennis build. Spend effort here in preference to a fourth agent.

8.4 Exchange handling

If Betfair is used as reference or as a price source, always convert to effective odds before comparison:

```
o_eff = 1 + (o − 1)(1 − c)
```

where `c` is the commission rate. Never compare a displayed exchange price to a bookmaker price.

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9. Schedule and publication

Tennis order of play, draws and withdrawals only firm up the night before, and withdrawals continue through the morning. The schedule below is a compromise between having current information and giving the audience time to act.

Time (UK)	Job
Rolling	Ingest results as matches complete
22:00	Ratings recompute. Deliberate gap after the day's play so completed-match data is settled and any injury or retirement reports have surfaced. Do not price a match minutes after the previous one ended.
22:00–23:30	Agents run. Manager runs. Price scanner sweeps.
23:30	Veto agent runs. Freshness gate checked.
00:00	Slate published for the following day.
07:00	Morning withdrawal sweep. Any affected selection is retracted and the retraction is published.
12:00	Second sweep covering matches starting after 16:00 UK.
At each match start	Closing price captured automatically.

Publication carries a standing note that selections are subject to the 07:00 check. Tennis withdrawals overnight are common enough that publishing at midnight without this is irresponsible.

Time zones. For Australian, Asian and any tournament outside European hours, the schedule shifts. Store every event's start time in UTC and derive the run schedule by working backwards from first serve for that tournament, not from a fixed UK clock.

Empty slates are a valid, expected output. Some days there is no bet. The publication layer must be able to say so plainly. If the system produces zero selections for seven consecutive days, the volume alert fires — that is more likely a bug than a quiet market.

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10. Backtest protocol

1. Load Sackmann `tennis_atp` history and tennis-data.co.uk odds. Join on player names — expect this to be genuinely painful and to need a name-normalisation table. Budget real time for it.
2. Walk-forward only. Train on years 1–N, test on N+1, roll.
3. Every feature carries `as_of`; the harness refuses to run if any feature's `as_of` post-dates the fixture.
4. **Minimum 300–500 settled selections before any result is discussed.** Below that the sample tells you nothing.
5. Report, in this order: calibration (reliability diagram and Brier by agent and by consensus), then CLV, then only after both, ROI.
6. **Record the number of model variants tested.** If ten configurations were tried and one looks excellent, that is roughly what pure noise produces. Without the count, skill and luck are indistinguishable.

10.1 Expected results, so you know whether you have a bug

- Agent 1 (Elo) Brier around .180–.190 — **worse than the market's .172**. Correct.
- Agent 2 (Markov) similar or slightly better on tour-level matches with good data.
- Agent 3 abstaining on the large majority of fixtures. Correct.
- Consensus firing on a small minority of the slate. Correct.
- Type B (price dispersion) flags substantially outnumbering Type A (model edge) flags. **Expected**, and it is the most informative single diagnostic in the build. If Type A dominates, your de-vig or your threshold is wrong.

If the backtest shows a large ROI on a small number of bets, look for the outlier before celebrating. That is exactly how the common-opponent result went wrong.